

Think — volume in context

The same volume, six different boxes: which one uses the least card?

LESSON 150

1 × 40 MIN

MATEMATIKA 6, O'YLAB KO'R

S7 15 IN CONTEXT

LESSON CLOCK

10'

10'

12'

8'

The packaging problem	10 min
Testing the boxes	10 min
What doubling does	12 min
Writing the report	8 min

LEARNING OBJECTIVES

- Find every cuboid with whole-number edges and a given volume.
- Compare their surface areas and identify the most economical.
- State what happens to volume and surface area when every edge is doubled.
- Turn the saving into a quantity of material and a cost.

TEXTBOOK

National	pp. 436–438
Cambridge	Stage 7, pp. 168–170
Workbook	Investigation 15

01

Explanation

The packaging problem

A factory must pack 216 cm³ of tea into a cuboid box with whole-number edges in centimetres. Every box holds the same tea. They do not use the same amount of card.

THE QUESTION

Among all cuboids of volume 216 cm³ with whole-number edges, which uses the least card — and is there a pattern that would let you answer the same question for any volume?

Start by listing boxes. Each one is three whole numbers whose product is 216.

Testing the boxes

BOX	VOLUME	THE THREE FACES	CARD USED
$6 \times 6 \times 6$	216 cm^3	36, 36, 36	216 cm^2
$9 \times 6 \times 4$	216 cm^3	54, 24, 36	228 cm^2
$8 \times 9 \times 3$	216 cm^3	72, 27, 24	246 cm^2
$12 \times 6 \times 3$	216 cm^3	72, 18, 36	252 cm^2
$12 \times 9 \times 2$	216 cm^3	108, 18, 24	300 cm^2
$216 \times 1 \times 1$	216 cm^3	216, 1, 216	866 cm^2

THE CONJECTURE

The nearer the box is to a cube, the less card it uses; the long thin box is by far the worst. The cube is the cheapest cuboid for any given volume.

What doubling does

Now change the size instead of the shape. Double every edge of a cube and both quantities grow — but not at the same rate.

CUBE	VOLUME	SURFACE AREA	CARD PER cm^3
edge 3 cm	27 cm^3	54 cm^2	2
edge 6 cm	216 cm^3	216 cm^2	1
edge 12 cm	1728 cm^3	864 cm^2	0.5

VOLUME GROWS FASTER THAN SURFACE AREA

Doubling the edges multiplies the volume by 8 and the surface area by only 4. That is why a large packet is cheaper per gram than a small one — and why a small animal loses heat faster than a large one.

Writing the report

Turn the finding into money, which is the form a factory understands.

QUANTITY	WORKING	ANSWER
card saved per box	$252 - 216$	36 cm^2
cost saved per box at 2 so‘m per cm^2	$36 \cdot 2$	72 so‘m
card saved on 10 000 boxes	$36 \cdot 10\,000$	36 m^2
money saved on 10 000 boxes	$72 \cdot 10\,000$	$720\,000\text{ so‘m}$

AND WHY IS THE CUBE NOT ALWAYS USED?

Because boxes must stand on shelves, be gripped by a hand, and hold a shape that is not itself a cube. A good report says what the mathematics recommends *and* what overrules it.

02 Worked examples

EXAMPLE 1 Compare the card used by a $12 \times 6 \times 3$ box and a $6 \times 6 \times 6$ box of the same volume.

- 1

Both hold 216 cm^3 .
- 2

$2(72 + 18 + 36) = 252\text{ cm}^2$ against $6 \cdot 36 = 216\text{ cm}^2$.
- 3

The cube saves 36 cm^2 per box.

About 14% less card ✓

ANSWER

252 against 216 cm^2

EXAMPLE 2 Find the card used by a $9 \times 6 \times 4$ box.

① The three faces: 54, 24, 36.

② $2(54 + 24 + 36)$.

③ $= 228 \text{ cm}^2$.

Between the cube and the $12 \times 6 \times 3$ box ✓

ANSWER

228 cm^2

EXAMPLE 3 A factory makes 10 000 boxes a day. How much card does the cube save, in square metres?

① 36 cm^2 per box.

② $36 \cdot 10\,000 = 360\,000 \text{ cm}^2$.

③ $360\,000 \div 10\,000 = 36 \text{ m}^2$.

1 m^2 is $10\,000 \text{ cm}^2$.

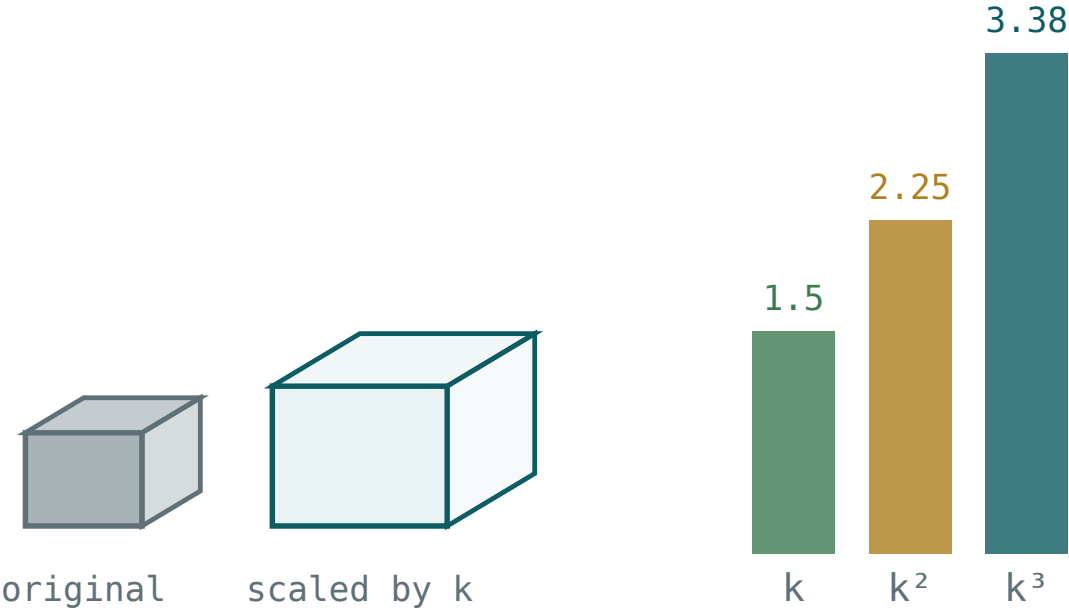
ANSWER

36 m^2 a day

03 Interactive model

Set $k = 2$ in the model and read the three multipliers aloud: 2, 4, 8. That single line is the whole of the third section.

Lengths, areas and volumes under a scale factor



k 1.5	length × 1.5	area ×k² 2.25	volume ×k³ 3.38
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04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Packaging	Qadoqlash	Упаковка
Economical	Tejamkor	Экономичный
Material	Material	Материал
Whole-number edge	Butun sonli qirra	Целое ребро
Scale factor	O'lchov koeffitsiyenti	Коэффициент подобия
To conjecture	Taxmin qilmoq	Выдвигать гипотезу
Saving	Tejamkorlik	Экономия
Per unit	Bir birlikka	На единицу

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Among cuboids of a fixed volume, the least card is used by:

the longest

the cube

the flattest

they are all equal

A $216 \times 1 \times 1$ box uses:

216 cm^2

433 cm^2

866 cm^2

218 cm^2

Doubling every edge multiplies the volume by:

2

4

6

8

It multiplies the surface area by:

2

4

6

8

Card per cm^3 for a large cube compared with a small one is:

larger

smaller

the same

zero

The cube is not always used because:

it is hard to make

shelves and contents have shapes too

it uses more card

it holds less

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The volume of a $6 \times 6 \times 6$ box

ANSWER 216 cm^3

2. The card used by that box

ANSWER 216 cm^2

3. The card used by a $12 \times 6 \times 3$ box

ANSWER 252 cm^2

4. The card used by a $9 \times 6 \times 4$ box

ANSWER 228 cm^2

5. Which of those three is the most economical?

ANSWER The cube

6. The volume of a $9 \times 6 \times 4$ box

ANSWER 216 cm^3

7. The card used by a $216 \times 1 \times 1$ box

ANSWER 866 cm^2

Medium · the standard question

7 PROBLEMS

1. The card used by an $8 \times 9 \times 3$ box

ANSWER 246 cm^2

2. The card used by a $12 \times 9 \times 2$ box

ANSWER 300 cm^2

3. The most economical shape for a fixed volume

ANSWER The cube

4. Order $6 \times 6 \times 6$, $9 \times 6 \times 4$, $12 \times 6 \times 3$ by card used

ANSWER $216, 228, 252 \text{ cm}^2$

5. The volume of a cube of edge 3 cm

ANSWER 27 cm^3

6. The surface area of a cube of edge 3 cm

ANSWER 54 cm^2

7. The surface area of a cube of edge 6 cm

ANSWER 216 cm^2

Hard · stretch and reason

7 PROBLEMS

1. Doubling every edge multiplies the volume by

ANSWER 8

2. ...and the surface area by

ANSWER 4

3. A cube of edge 10 cm: its volume and surface area

ANSWER 1000 cm³ and 600 cm²

4. Card per cm³ for cubes of edge 3 and 6

ANSWER 2 against 1

5. At 2 so'm per cm², the cube saves against $12 \times 6 \times 3$

ANSWER 72 so'm a box

6. On 10 000 boxes, the card saved

ANSWER 36 m²

7. Why is a cube not always used?

ANSWER Shelves, handling and the shape of the contents

07 Homework

Homework — the report

One page: the table of boxes, the conjecture, the doubling result, and the cost in so'm.

1. List every cuboid with whole-number edges and volume 64 cm³, and find the card each uses.
2. State which is the most economical and whether it agrees with the conjecture.
3. Work out what happens to the volume and the surface area of a cube when every edge is trebled.
4. A factory makes 5000 boxes a day. Find the card saved by the best shape, in m².

5. Write two sentences on why a factory might still not choose the best shape.